KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5432 | 5 Credits: 4 | Program Core | Source-grounded

Automation and Control

 To identify the significance of automation.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5432 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5432.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Mechatronics
Course code	5432
Course title	Automation and Control
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3 T:1 P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To identify the significance of automation.
- To provide a basic knowledge about the structure and application of power
- semiconductor devices.
- To develop PLC ladder programs for real time applications.

Course outcomes

CO	Official outcome	Study level
CO1	13 Understanding and its applications Illustrate the structure and operation of Power	See official syllabus
CO2	16 Understanding Semiconductor Devices. Explain the working of Controlled Rectifiers and	See official syllabus
CO3	16 Understanding Inverters.	See official syllabus
CO4	Develop PLC ladder program for an application. 13 Applying	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Understand the concept of Industrial Automation and its applications	4	As specified
Module 2	Illustrate the structure and operation of Power Semiconductor Devices	4	As specified
Module 3	Explain the working of Controlled Rectifiers and Inverters	3	As specified
Module 4	Develop PLC ladder program for an application	4	As specified

MODULE 1

Understand the concept of Industrial Automation and its applications

This module is presented from the official syllabus scope for Automation and Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the concept of Industrial Automation and its applications
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Describe the need for automation

Describe the need for automation is an official topic in Module 1 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the need for automation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the need for automation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the need for automation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Describe the need for automation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the types of automation

Explain the types of automation is an official topic in Module 1 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the types of automation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the types of automation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the types of automation means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the types of automation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the elements of an automated system

Explain the elements of an automated system is an official topic in Module 1 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the elements of an automated system using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the elements of an automated system should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the elements of an automated system means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Explain the elements of an automated system [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the levels of automation 4 Understanding Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation.

Illustrate the levels of automation 4 Understanding Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation. is an official topic in Module 1 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the levels of automation 4 Understanding Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the levels of automation 4 understanding automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the levels of automation 4 Understanding Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Illustrate the levels of automation 4 Understanding Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Understand the concept of Industrial Automation and its applications

   M1.01      Describe the need for automation                    3
Understanding

   M1.02      Explain the types of automation                     3
Understanding

   M1.03      Explain the elements of an automated system         3
Understanding

   M1.04      Illustrate the levels of automation                 4
Understanding

 Contents:
 Automation - need for automation - advantages and disadvantages of automation –
ten
 strategies for automation - types of automation - fixed, programmable, flexible
and
 integrated automation – advantages and disadvantages - basic elements of an
automated
 system – levels of automation - future challenges of automation.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Illustrate the structure and operation of Power Semiconductor Devices

This module is presented from the official syllabus scope for Automation and Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Illustrate the structure and operation of Power Semiconductor Devices
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding SCR

4 Understanding SCR is an official topic in Module 2 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding SCR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding scr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding SCR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding SCR □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Explain the turn-on methods for SCR

Explain the turn-on methods for SCR is an official topic in Module 2 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the turn-on methods for SCR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the turn-on methods for scr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the turn-on methods for SCR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain the turn-on methods for SCR [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the commutation techniques of SCR

Illustrate the commutation techniques of SCR is an official topic in Module 2 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the commutation techniques of SCR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the commutation techniques of scr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the commutation techniques of SCR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Illustrate the commutation techniques of SCR [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the structure and characteristics of 4 Understanding DIAC and TRIAC SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications.

Illustrate the structure and characteristics of 4 Understanding DIAC and TRIAC SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications. is an official topic in Module 2 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the structure and characteristics of 4 Understanding DIAC and TRIAC SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the structure and characteristics of 4 understanding diac and triac scr - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘r’ triggering - rc triggering - ujt triggering - commutation techniques - forced commutation circuits (class a to e) diac, triac- structure - working principle – characteristics – applications.

should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the structure and characteristics of 4 Understanding DIAC and TRIAC SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate the structure and characteristics of 4 Understanding DIAC and TRIAC SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Illustrate the structure and operation of Power Semiconductor Devices

              Illustrate the structure and characteristics of
   M2.01                                                          4
Understanding
              SCR

  M2.02       Explain the turn-on methods for SCR                 4
Understanding

   M2.03      Illustrate the commutation techniques of SCR        4
Understanding

   M2.04      Illustrate the structure and characteristics of     4
Understanding
              DIAC and TRIAC
              Series Test 1                                       1
Contents:
SCR - structure - characteristics - working principle – applications - two
transistor analogy
– turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC
triggering -
UJT triggering - commutation techniques - forced commutation circuits (class A to
E)
DIAC, TRIAC- structure - working principle – characteristics – applications.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the working of Controlled Rectifiers and Inverters

This module is presented from the official syllabus scope for Automation and Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the working of Controlled Rectifiers and Inverters
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Illustrate the working of single-phase controlled 6 Understanding rectifiers

Illustrate the working of single-phase controlled 6 Understanding rectifiers is an official topic in Module 3 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the working of single-phase controlled 6 Understanding rectifiers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the working of single-phase controlled 6 understanding rectifiers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the working of single-phase controlled 6 Understanding rectifiers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

> **Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ Illustrate the working of single-phase controlled 6 Understanding rectifiers □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Describe the operation of inverter circuits

Describe the operation of inverter circuits is an official topic in Module 3 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the operation of inverter circuits using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the operation of inverter circuits should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the operation of inverter circuits means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Describe the operation of inverter circuits [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain the working of single-phase dual 4 Understanding converter Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications.

Explain the working of single-phase dual 4 Understanding converter Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications. is an official topic in Module 3 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the working of single-phase dual 4 Understanding converter Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the working of single-phase dual 4 understanding converter single phase converters - half wave - full wave midpoint and bridge - working principle with r, rl loads - basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Explain the working of single-phase dual 4 Understanding converter Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain the working of single-phase dual 4 Understanding converter Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the working of Controlled Rectifiers and Inverters

  M3.01      Illustrate the working of single-phase controlled       6
Understanding
             rectifiers
  M3.02      Describe the operation of inverter circuits             6
Understanding

  M3.03      Explain the working of single-phase dual                4
Understanding
             converter

Contents:
Single phase converters - half wave - full wave midpoint and bridge - working
principle
with R, RL loads - Basic inverter circuit - working principle - series and parallel
inverter
circuits - working principle – single phase dual converters - working principle -
applications.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Develop PLC ladder program for an application

This module is presented from the official syllabus scope for Automation and Control. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop PLC ladder program for an application
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the architecture of PLC

Explain the architecture of PLC is an official topic in Module 4 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the architecture of PLC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the architecture of plc should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the architecture of PLC means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain the architecture of PLC □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

List the advantages and applications of PLC 2 Understanding Demonstrate ladder diagram instruction sets

List the advantages and applications of PLC 2 Understanding Demonstrate ladder diagram instruction sets is an official topic in Module 4 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify List the advantages and applications of PLC 2 Understanding Demonstrate ladder diagram instruction sets using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, list the advantages and applications of plc 2 understanding demonstrate ladder diagram instruction sets should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What List the advantages and applications of PLC 2 Understanding Demonstrate ladder diagram instruction sets means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ List the advantages and applications of PLC 2 Understanding Demonstrate ladder diagram instruction sets □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

4 Understanding of PLC. Develop ladder program for various

4 Understanding of PLC. Develop ladder program for various is an official topic in Module 4 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding of PLC. Develop ladder program for various using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding of plc. develop ladder program for various should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding of PLC. Develop ladder program for various means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding of PLC. Develop ladder program for various □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

4 Applying applications PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. Text / Reference: T/R Book Title/ Author T1 Industrial automation and mechatronics - A.K.Gupta, S.K.Arora T2 Industrial Electronics and Control - S K Bhattacharya, S Chatterjee T3 Programmable logic controllers - Frank D Petruzella. R1 Industrial Electronics and Control - Biswanath Paul - PHI R2 Power Electronics - P S Bimbhra Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - R3 Delmar Online Resources: Sl No Website Link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/

4 Applying applications PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. Text / Reference: T/R Book Title/ Author T1 Industrial automation and mechatronics - A.K.Gupta, S.K.Arora T2 Industrial Electronics and Control - S K Bhattacharya, S Chatterjee T3 Programmable logic controllers - Frank D Petruzella. R1 Industrial Electronics and Control - Biswanath Paul - PHI R2 Power Electronics - P S Bimbhra Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - R3 Delmar Online Resources: Sl No Website Link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/ is an official topic in Module 4 of Automation and Control. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying applications PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. Text / Reference: T/R Book Title/ Author T1 Industrial automation and mechatronics - A.K.Gupta, S.K.Arora T2 Industrial Electronics and Control - S K Bhattacharya, S Chatterjee T3 Programmable logic controllers - Frank D Petruzella. R1 Industrial Electronics and Control - Biswanath Paul - PHI R2 Power Electronics - P S Bimbhra Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - R3 Delmar Online Resources: Sl No Website Link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying applications plc - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. text / reference: t/r book title/ author t1 industrial automation and mechatronics - a.k.gupta, s.k.arora t2 industrial electronics and control - s k bhattacharya, s chatterjee t3 programmable logic controllers - frank d petruzella. r1 industrial electronics and control - biswanath paul - phi r2 power electronics - p s bimbhra introduction to programmable logic controllers - gary dunning - 3rd edition - r3 delmar online resources: sl no website link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying applications PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. Text /

Aspect	What to prepare
	Reference: T/R Book Title/ Author T1 Industrial automation and mechatronics - A.K.Gupta, S.K.Arora T2 Industrial Electronics and Control - S K Bhattacharya, S Chatterjee T3 Programmable logic controllers - Frank D Petruzella. R1 Industrial Electronics and Control - Biswanath Paul - PHI R2 Power Electronics - P S Bimbhra Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - R3 Delmar Online Resources: Sl No Website Link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Applying applications PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. Text / Reference: T/R Book Title/ Author T1 Industrial automation and mechatronics - A.K.Gupta, S.K.Arora T2 Industrial Electronics and Control - S K Bhattacharya, S Chatterjee T3 Programmable logic controllers - Frank D Petruzella. R1 Industrial Electronics and Control - Biswanath Paul - PHI R2 Power Electronics - P S Bimbhra Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - R3 Delmar Online Resources: Sl No Website Link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/ □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Develop PLC ladder program for an application

  M4.01      Explain the architecture of PLC                        3
Understanding

  M4.02      List the advantages and applications of PLC            2
Understanding
             Demonstrate ladder diagram instruction sets
  M4.03                                                             4
Understanding
             of PLC.
             Develop ladder program for various
  M4.04                                                             4
Applying
             applications

             Series Test                                            1

Contents:
PLC - architecture - advantages - applications - ladder logic - ladder diagram
instruction
sets - bit instructions - timer/counter instructions - compare instructions -
move instructions
- math instructions - program control instructions - ladder programs – logic
gates - staircase
lamp – conveyor control – water level control – servo motor speed control.
Text / Reference:
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Describe the need for automation. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the need for automation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the types of automation. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the types of automation*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the elements of an automated system. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the elements of an automated system*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate the levels of automation 4 Understanding Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation.. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the levels of automation 4 Understanding Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation.*. State the governing principle, list the key components or stages, and close with one relevant application or observation.

stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding SCR. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding SCR*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain the turn-on methods for SCR. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the turn-on methods for SCR*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate the commutation techniques of SCR. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the commutation techniques of SCR*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Illustrate the structure and characteristics of 4 Understanding DIAC and TRIAC SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications.. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the structure and characteristics of 4 Understanding DIAC and TRIAC SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 3

Define or explain Illustrate the working of single-phase controlled 6 Understanding rectifiers. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the working of single-phase controlled 6 Understanding rectifiers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain Describe the operation of inverter circuits. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the operation of inverter circuits*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain Explain the working of single-phase dual 4 Understanding converter Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the working of single-phase dual 4 Understanding converter Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Module 4

Define or explain Explain the architecture of PLC. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the architecture of PLC*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

Define or explain List the advantages and applications of PLC 2 Understanding Demonstrate ladder diagram instruction sets. (3 marks)

Answer: Begin with the standard definition or identity of *List the advantages and applications of PLC 2 Understanding Demonstrate ladder diagram instruction sets*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding of PLC. Develop ladder program for various. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding of PLC. Develop ladder program for various*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Applying applications PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. Text / Reference: T/R Book Title/ Author T1 Industrial automation and mechatronics - A.K.Gupta, S.K.Arora T2 Industrial Electronics and Control - S K Bhattacharya, S Chatterjee T3 Programmable logic controllers - Frank D Petruzella. R1 Industrial Electronics and Control - Biswanath Paul - PHI R2 Power Electronics - P S Bimbhra Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - R3 Delmar Online Resources: Sl No Website Link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/. (3 marks)

Answer: Begin with the standard definition or identity of *4 Applying applications PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control. Text / Reference: T/R Book Title/ Author T1 Industrial automation and mechatronics - A.K.Gupta, S.K.Arora T2 Industrial Electronics and Control - S K Bhattacharya, S Chatterjee T3 Programmable logic controllers - Frank D Petruzella. R1 Industrial Electronics and Control - Biswanath Paul - PHI R2 Power Electronics - P S Bimbhra Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - R3 Delmar Online Resources: Sl No Website Link 1 https://nptel.ac.in/courses/108105062 2 https://nptel.ac.in/courses/108105063 3 https://www.electrical4u.com/electrical-engineering-articles/power-electronics/ 4 https://nptel.ac.in/courses/108/102/108102145/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Describe the need for automation**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding SCR**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the working of single-phase controlled 6 Understanding rectifiers**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain the architecture of PLC**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.

- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl No Website Link https://nptel.ac.in/courses/108105062 https://nptel.ac.in/courses/108105063
- https://www.electrical4u.com/electrical-engineering-articles/power-electronics/
- https://nptel.ac.in/courses/108/102/108102145/

IN-PAGE SEARCH

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